Lab4_gender_edu

2026-10-05

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.2.1     ✔ readr     2.2.0
## ✔ forcats   1.0.1     ✔ stringr   1.6.0
## ✔ ggplot2   4.0.3     ✔ tibble    3.3.1
## ✔ lubridate 1.9.5     ✔ tidyr     1.3.2
## ✔ purrr     1.2.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

```
library(AER)
```

```
## Loading required package: car
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
##
## Loading required package: lmtest
## Loading required package: zoo
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
##
## Loading required package: sandwich
## Loading required package: survival
```

```
load("d_HHP2020_24.RData")
prime_age_laborforce_data <- d_HHP2020_24 %>% filter(!is.na(work_kind) &
                                                       Age >= 25 & Age <= 55)
```

Check gender categories

```
table(prime_age_laborforce_data$Gender)
```

```
##
##   male female  trans  other
## 166600 221034    948   2358
```

```
d <- subset(prime_age_laborforce_data,
            Gender %in% c("male", "female") & Education != "lt hs")
d$Education <- factor(d$Education,
                      levels = c("some hs", "high school", "some college",
                                 "assoc deg", "college grad", "adv degree"))
d$Gender <- factor(d$Gender, levels = c("male", "female"))
table(d$Gender, d$Education)
```

```
##
##          some hs high school some college assoc deg college grad adv degree
##   male      1473       14795        29955     14267        58260      47151
##   female    1918       18303        38448     23688        70913      66998
```

Looking to see if the gender gap (between men and women) changes as education is increased

```
model_a <- lm(income_midpoint ~ Age + Gender * Education, data = d)
summary(model_a)
```

```
##
## Call:
## lm(formula = income_midpoint ~ Age + Gender * Education, data = d)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -158372  -44067   -8847   44430  198748
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         10107.95    1865.98   5.417 6.07e-08 ***
## Age                                  1237.95      12.58  98.419  < 2e-16 ***
## Genderfemale                       -20995.07    2388.45  -8.790  < 2e-16 ***
## Educationhigh school                17076.25    1871.10   9.126  < 2e-16 ***
## Educationsome college               33323.87    1827.78  18.232  < 2e-16 ***
## Educationassoc deg                  34191.24    1868.90  18.295  < 2e-16 ***
## Educationcollege grad               71744.47    1808.48  39.671  < 2e-16 ***
## Educationadv degree                 92676.34    1812.43  51.134  < 2e-16 ***
## Genderfemale:Educationhigh school     781.69    2498.89   0.313    0.754
## Genderfemale:Educationsome college  -1884.30    2440.40  -0.772    0.440
## Genderfemale:Educationassoc deg      3463.19    2484.78   1.394    0.163
## Genderfemale:Educationcollege grad    362.65    2414.71   0.150    0.881
## Genderfemale:Educationadv degree     1500.29    2418.15   0.620    0.535
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 58660 on 324918 degrees of freedom
##   (61238 observations deleted due to missingness)
## Multiple R-squared:  0.2132, Adjusted R-squared:  0.2132
## F-statistic:  7338 on 12 and 324918 DF,  p-value: < 2.2e-16
```

The standard error is bigger than all but 1 of the estimates. T values suggest there is mostly noise in these numbers. Of possible interest is the result for female:Associate degree. This produced the highest reduction in wage gap but doesn't give a high enough confidence interval to make it meaningful. The one conclusion we can make is that there is approximately a $20000 wage gap at all education levels.

```
model_b <- lm(income_midpoint ~ Age + Gender * Education + Race, data = d)
summary(model_b)
```

```
##
## Call:
## lm(formula = income_midpoint ~ Age + Gender * Education + Race,
##     data = d)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -168773  -43504   -8297   43788  205314
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        12103.51    1853.58   6.530 6.60e-11 ***
## Age                                 1250.59      12.49 100.112  < 2e-16 ***
## Genderfemale                      -19141.48    2370.78  -8.074 6.83e-16 ***
## Educationhigh school               16758.78    1857.26   9.023  < 2e-16 ***
## Educationsome college              32841.21    1814.28  18.101  < 2e-16 ***
## Educationassoc deg                 33436.52    1855.10  18.024  < 2e-16 ***
## Educationcollege grad              69856.89    1795.25  38.912  < 2e-16 ***
## Educationadv degree                90295.61    1799.37  50.182  < 2e-16 ***
## RaceBlack                         -24550.41     397.88 -61.703  < 2e-16 ***
## RaceAsian                          10090.82     429.23  23.509  < 2e-16 ***
## Raceother                          -9940.08     458.14 -21.696  < 2e-16 ***
## Genderfemale:Educationhigh school    -60.25    2480.25  -0.024    0.981
## Genderfemale:Educationsome college  -2346.68    2422.21  -0.969    0.333
## Genderfemale:Educationassoc deg      2752.57    2466.24   1.116    0.264
## Genderfemale:Educationcollege grad   -585.34    2396.71  -0.244    0.807
## Genderfemale:Educationadv degree     1266.47    2400.12   0.528    0.598
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 58220 on 324915 degrees of freedom
##   (61238 observations deleted due to missingness)
## Multiple R-squared:  0.225,  Adjusted R-squared:  0.2249
## F-statistic:  6288 on 15 and 324915 DF,  p-value: < 2.2e-16
```

Looking to see if race affects the wage gap, first advanced degrees

```
adv <- subset(d, Education == "adv degree")
model_d <- lm(income_midpoint ~ Age + Gender * Race, data = adv)
summary(model_d)
```

```
##
## Call:
## lm(formula = income_midpoint ~ Age + Gender * Race, data = adv)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -172621  -51993   -2944   57616  138117
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              94452.08    1145.22  82.475  < 2e-16 ***
## Age                       1430.03      25.93  55.156  < 2e-16 ***
## Genderfemale            -17988.73     454.14 -39.611  < 2e-16 ***
## RaceBlack               -20509.86    1493.99 -13.728  < 2e-16 ***
## RaceAsian                12017.03     916.32  13.114  < 2e-16 ***
## Raceother               -11096.31    1623.18  -6.836 8.18e-12 ***
## Genderfemale:RaceBlack   -4821.34    1767.46  -2.728  0.00638 **
## Genderfemale:RaceAsian    7494.13    1370.36   5.469 4.54e-08 ***
## Genderfemale:Raceother    2227.01    2053.65   1.084  0.27818
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 62620 on 99599 degrees of freedom
##   (14541 observations deleted due to missingness)
## Multiple R-squared:  0.06652,    Adjusted R-squared:  0.06645
## F-statistic: 887.2 on 8 and 99599 DF,  p-value: < 2.2e-16
```

For advanced degrees, it appears that being Black and being Asian both provide meaningful effects. The gap widens by $4800 for Black women (t value -2.7) and shrinks by almost $7500 for Asian women (t value 5.5). These findings have a low R squared, however, suggesting that they only explain <7% of the difference in wages. Race 'Other' has a t value that is too low to be significant.

```
cg <- subset(d, Education == "college grad")
model_cg <- lm(income_midpoint ~ Age + Gender * Race, data = cg)
summary(model_cg)
```

```
##
## Call:
## lm(formula = income_midpoint ~ Age + Gender * Race, data = cg)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -145198  -49390   -6890   48291  161610
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              80666.31     936.00  86.182  < 2e-16 ***
## Age                       1295.15      22.08  58.648  < 2e-16 ***
## Genderfemale            -20009.61     409.19 -48.901  < 2e-16 ***
## RaceBlack               -25909.88    1426.07 -18.169  < 2e-16 ***
## RaceAsian                 5798.57    1027.58   5.643 1.68e-08 ***
## Raceother               -12153.88    1339.15  -9.076  < 2e-16 ***
## Genderfemale:RaceBlack   -3735.84    1749.09  -2.136   0.0327 *
## Genderfemale:RaceAsian    6160.31    1490.11   4.134 3.57e-05 ***
## Genderfemale:Raceother    2607.59    1777.15   1.467   0.1423
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 61650 on 110197 degrees of freedom
##   (18967 observations deleted due to missingness)
## Multiple R-squared:  0.06737,    Adjusted R-squared:  0.0673
## F-statistic:   995 on 8 and 110197 DF,  p-value: < 2.2e-16
```

We get similar results with College degree. The gap widens for Black women by $3736, and narrows by $6160 for Asian women with similar R squared.

```
ad <- subset(d, Education == "assoc deg")
model_ad <- lm(income_midpoint ~ Age + Gender * Race, data = ad)
summary(model_ad)
```

```
##
## Call:
## lm(formula = income_midpoint ~ Age + Gender * Race, data = ad)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -99120 -35949 -11400  31732 174029
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             55268.22    1549.97  35.658  < 2e-16 ***
## Age                      1024.58      35.34  28.993  < 2e-16 ***
## Genderfemale           -16204.43     651.47 -24.874  < 2e-16 ***
## RaceBlack              -18880.73    1957.01  -9.648  < 2e-16 ***
## RaceAsian               -6596.37    2526.52  -2.611  0.00904 **
## Raceother              -11502.17    1936.35  -5.940 2.88e-09 ***
## Genderfemale:RaceBlack  -8146.17    2296.39  -3.547  0.00039 ***
## Genderfemale:RaceAsian   9571.19    3413.19   2.804  0.00505 **
## Genderfemale:Raceother   1665.81    2447.44   0.681  0.49611
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 50710 on 31191 degrees of freedom
##   (6755 observations deleted due to missingness)
## Multiple R-squared:  0.06937,    Adjusted R-squared:  0.06913
## F-statistic: 290.6 on 8 and 31191 DF,  p-value: < 2.2e-16
```

The effect gets more pronounced with Associate degrees. The wage gap increases by $8146 for Black women and decreases by $9571 for Asian women (more than half of the baseline gap). The t values suggest significant results but the R squared remains just under 7%.

```
sc <- subset(d, Education == "some college")
model_sc <- lm(income_midpoint ~ Age + Gender * Race, data = sc)
summary(model_sc)
```

```
##
## Call:
## lm(formula = income_midpoint ~ Age + Gender * Race, data = sc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -101790  -37222  -11801   28381  192903
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              44917.09    1137.69  39.481  < 2e-16 ***
## Age                       1261.33      26.14  48.259  < 2e-16 ***
## Genderfemale            -22331.11     493.16 -45.281  < 2e-16 ***
## RaceBlack               -21726.00    1314.97 -16.522  < 2e-16 ***
## RaceAsian                -4477.26    1925.73  -2.325  0.02008 *
## Raceother               -12398.66    1346.14  -9.211  < 2e-16 ***
## Genderfemale:RaceBlack    -296.16    1603.89  -0.185  0.85350
## Genderfemale:RaceAsian   11862.67    2736.53   4.335 1.46e-05 ***
## Genderfemale:Raceother    4978.60    1754.31   2.838  0.00454 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 52000 on 55625 degrees of freedom
##   (12769 observations deleted due to missingness)
## Multiple R-squared:  0.09624,    Adjusted R-squared:  0.09611
## F-statistic: 740.4 on 8 and 55625 DF,  p-value: < 2.2e-16
```

For ‘some college’ there doesn’t appear to be a reliable result for Black women. Asian women, however close the gap by $11862. R squared is now 9.6%.

```
hs <- subset(d, Education == "high school")
model_hs <- lm(income_midpoint ~ Age + Gender * Race, data = hs)
summary(model_hs)
```

```
##
## Call:
## lm(formula = income_midpoint ~ Age + Gender * Race, data = hs)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -79885 -31357  -9687  18520 198456
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              46719.63    1466.97  31.848  < 2e-16 ***
## Age                        830.27      33.47  24.803  < 2e-16 ***
## Genderfemale           -19367.59     640.63 -30.232  < 2e-16 ***
## RaceBlack              -21897.79    1665.79 -13.146  < 2e-16 ***
## RaceAsian              -10644.81    2625.62  -4.054 5.05e-05 ***
## Raceother              -10938.44    1707.31  -6.407 1.51e-10 ***
## Genderfemale:RaceBlack  -2157.83    2069.51  -1.043    0.297
## Genderfemale:RaceAsian   6132.53    3764.28   1.629    0.103
## Genderfemale:Raceother   2544.84    2302.20   1.105    0.269
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 46190 on 25825 degrees of freedom
##   (7264 observations deleted due to missingness)
## Multiple R-squared:  0.08559,    Adjusted R-squared:  0.08531
## F-statistic: 302.2 on 8 and 25825 DF,  p-value: < 2.2e-16
```

t values of ‘HS grad’ are too low to provide reliable results

```
shs <- subset(d, Education == "some hs")
model_shs <- lm(income_midpoint ~ Age + Gender * Race, data = shs)
summary(model_shs)
```

```
##
## Call:
## lm(formula = income_midpoint ~ Age + Gender * Race, data = shs)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -59762 -27162  -9541  13999 196907
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)              43666.1     4664.7   9.361  < 2e-16 ***
## Age                        519.9      105.4   4.935 8.56e-07 ***
## Genderfemale           -22201.6     1975.2 -11.240  < 2e-16 ***
## RaceBlack              -15173.4     4478.6  -3.388 0.000715 ***
## RaceAsian              -12510.1     5058.8  -2.473 0.013467 *
## Raceother              -10348.1     4251.3  -2.434 0.015000 *
## Genderfemale:RaceBlack    484.8     5455.9   0.089 0.929204
## Genderfemale:RaceAsian  22770.3     8406.5   2.709 0.006803 **
## Genderfemale:Raceother   2543.0     5975.4   0.426 0.670457
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41620 on 2440 degrees of freedom
##   (942 observations deleted due to missingness)
## Multiple R-squared:  0.08965,    Adjusted R-squared:  0.08666
## F-statistic: 30.04 on 8 and 2440 DF,  p-value: < 2.2e-16
```

The t value for black women is too low to suggest a detectable difference. Asian women with 'some HS' appear to completely close the gender wage gap.

```
model_c <- lm(income_midpoint ~ Age + Gender + Race * Education, data = d)

new <- expand.grid(Age = 40,
                   Gender = "female",
                   Race = c("white", "Black", "Asian", "other"),
                   Education = levels(d$Education))
new$yhat <- predict(model_c, newdata = new)

tab <- round(xtabs(yhat ~ Race + Education, data = new))
tab <- as.data.frame.matrix(tab)
tab <- tab[c("white", "Black", "Asian", "other"), levels(d$Education)]

men <- data.frame(Age = 40, Gender = "male", Race = "white",
                  Education = levels(d$Education))
tab["white men", ] <- round(predict(model_c, newdata = men))

tab[] <- lapply(tab, function(x) paste0("$", format(x, big.mark = ",", trim = TRUE)))
rownames(tab)[1:4] <- c("white women", "Black women", "Asian women", "other women")
knitr::kable(tab, align = "r", caption = "Predicted income, age 40")
```

Predicted income, age 40

	some hs	high school	some college	assoc deg	college grad	adv degree
white women	$42,414	$59,982	$74,547	$78,534	$112,894	$133,522
Black women	$27,046	$36,681	$52,151	$53,921	$84,410	$109,853
Asian women	$37,653	$52,446	$76,085	$76,898	$121,663	$148,509
other women	$33,797	$51,011	$64,923	$68,308	$102,159	$123,759
white men	$61,479	$79,047	$93,612	$97,599	$131,959	$152,586

```
confint(model_a)
```

```
##                                       2.5 %     97.5 %
## (Intercept)                        6450.688  13765.214
## Age                                1213.301   1262.608
## Genderfemale                     -25676.358 -16313.783
## Educationhigh school              13408.941  20743.558
## Educationsome college             29741.465  36906.275
## Educationassoc deg                30528.251  37854.232
## Educationcollege grad             68199.902  75289.038
## Educationadv degree               89124.021  96228.651
## Genderfemale:Educationhigh school   -4116.056   5679.435
## Genderfemale:Educationsome college  -6667.415   2898.810
## Genderfemale:Educationassoc deg     -1406.902   8333.288
## Genderfemale:Educationcollege grad  -4370.114   5095.405
## Genderfemale:Educationadv degree    -3239.212   6239.790
```

The 95% interval for gender female excludes 0, and is thus, significant. The intervals for education level all include 0, and thus the gender gap does not differ detectably across education levels.

```
anova(lm(income_midpoint ~ Age + Gender + Education, data = d), model_a)
```

```
## Analysis of Variance Table
##
## Model 1: income_midpoint ~ Age + Gender + Education
## Model 2: income_midpoint ~ Age + Gender * Education
##   Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
## 1 324923 1.1181e+15
## 2 324918 1.1179e+15  5 1.6613e+11 9.6571 3.111e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F = 9.66, p <.001, suggesting that the gender gap does vary by education level, contradicting the confidence interval findings, though the improvement in fit is small.

```
library(AER)
coeftest(model_a, vcov = vcovHC(model_a, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                                    Estimate Std. Error  t value  Pr(>|t|)
## (Intercept)                       10107.951   1564.651   6.4602 1.047e-10 ***
## Age                                1237.954     12.374 100.0410 < 2.2e-16 ***
## Genderfemale                     -20995.071   1781.106 -11.7877 < 2.2e-16 ***
## Educationhigh school              17076.250   1553.294  10.9936 < 2.2e-16 ***
## Educationsome college             33323.870   1520.800  21.9121 < 2.2e-16 ***
## Educationassoc deg                34191.242   1557.236  21.9564 < 2.2e-16 ***
## Educationcollege grad             71744.470   1505.139  47.6663 < 2.2e-16 ***
## Educationadv degree               92676.336   1510.495  61.3549 < 2.2e-16 ***
## Genderfemale:Educationhigh school   781.689   1878.794   0.4161   0.67737
## Genderfemale:Educationsome college -1884.303   1838.134  -1.0251   0.30531
## Genderfemale:Educationassoc deg     3463.193   1881.897   1.8403   0.06573 .
## Genderfemale:Educationcollege grad   362.646   1820.522   0.1992   0.84211
## Genderfemale:Educationadv degree    1500.289   1826.855   0.8212   0.41151
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Standard errors shrank 20%-25%. T values remained significant for Education and Gender but failed to show an impact on wage gap across educational levels. Associate degree t increased to 1.84 making it significant at the 10% level but not at 5%.

```
model_log <- lm(log(income_midpoint) ~ Age + Gender * Education, data = d)
summary(model_log)
```

```
##
## Call:
## lm(formula = log(income_midpoint) ~ Age + Gender * Education,
##     data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.56353 -0.39554  0.09411  0.48681  2.18441
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      10.19151    0.02076 490.842  < 2e-16 ***
## Age                               0.01338    0.00014  95.626  < 2e-16 ***
## Genderfemale                     -0.45359    0.02658 -17.067  < 2e-16 ***
## Educationhigh school              0.29347    0.02082  14.095  < 2e-16 ***
## Educationsome college             0.49989    0.02034  24.579  < 2e-16 ***
## Educationassoc deg                0.53805    0.02080  25.873  < 2e-16 ***
## Educationcollege grad             0.89901    0.02012  44.675  < 2e-16 ***
## Educationadv degree               1.06937    0.02017  53.024  < 2e-16 ***
## Genderfemale:Educationhigh school   0.11935    0.02781   4.292 1.77e-05 ***
## Genderfemale:Educationsome college  0.13321    0.02716   4.906 9.32e-07 ***
## Genderfemale:Educationassoc deg     0.21119    0.02765   7.638 2.21e-14 ***
## Genderfemale:Educationcollege grad  0.25278    0.02687   9.408  < 2e-16 ***
## Genderfemale:Educationadv degree    0.29930    0.02691  11.123  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6527 on 324918 degrees of freedom
##   (61238 observations deleted due to missingness)
## Multiple R-squared:  0.2174, Adjusted R-squared:  0.2174
## F-statistic:  7522 on 12 and 324918 DF,  p-value: < 2.2e-16
```

The t values for genderfemale:education level become significant in the log model. While the absolute $value of the gap remained around $20000, the percentage of the gap falls as incomes rise.

```
b <- coef(model_log)
gap <- c(b["Genderfemale"],
         b["Genderfemale"] + b[grep("Genderfemale:", names(b))])
names(gap) <- levels(d$Education)
round(100 * (exp(gap) - 1), 1)
```

```
##      some hs  high school some college    assoc deg college grad   adv degree
##        -36.5        -28.4        -27.4        -21.5        -18.2        -14.3
```

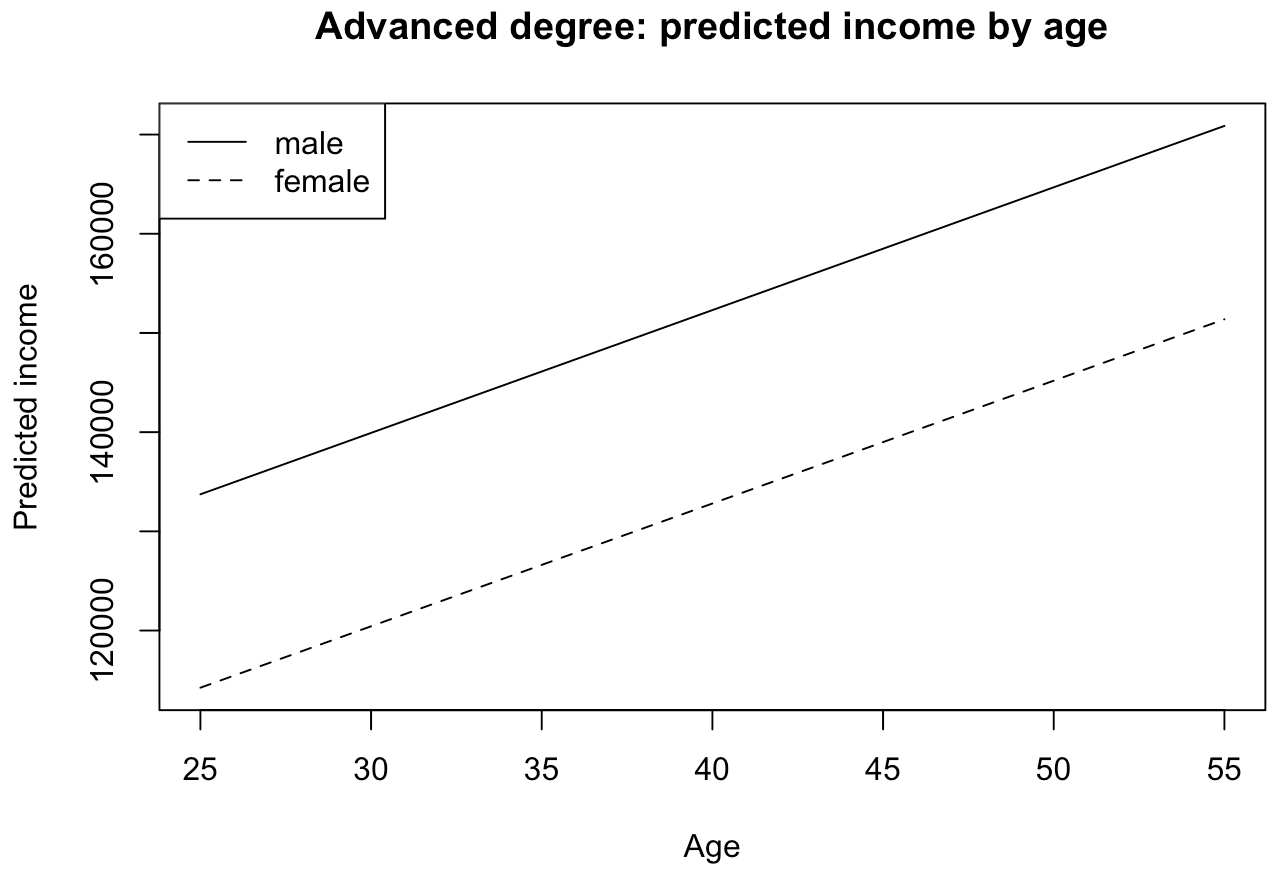
```
mean(fitted(model_a))
```

```
## [1] 112450.5
```

```
mean(exp(fitted(model_log)))
```

```
## [1] 95066.05
```

```
ages <- 25:55
pm <- predict(model_a, data.frame(Age = ages, Gender = "male", Education = "adv degre
e"))
pf <- predict(model_a, data.frame(Age = ages, Gender = "female", Education = "adv degre
e"))
plot(ages, pm, type = "l", ylim = range(pm, pf),
     xlab = "Age", ylab = "Predicted income",
     main = "Advanced degree: predicted income by age")
lines(ages, pf, lty = 2)
legend("topleft", c("male", "female"), lty = 1:2)
```



These results show associations, not proven causes. Education plausibly affects income, given more schooling presumably increases skills and makes one more valuable to an employer. It is clear that there is a difference in income by gender, with an approximately $20000 gap between men and women at each educational level. This gap remains roughly the same as incomes rise so decreases in proportion at higher income and educational levels. This is demonstrated in the log model. The low R squared (7% to 22%) shows that most of what drives income is outside of this model, such as hours worked, field of study, years of experience, and region.

I restricted the data set to working age people to avoid students and retirees. I dropped ‘it hs’ as this was a very small number of people. I ran gender-by-race model separately for each educational level, so race and gender gaps could differ by education. Hispanic ethnicity was not controlled in the education level models. Adding it to the race by education model left the gender gap unchanged but reduced the Asian and Other gaps at lower educational levels.

```
model_c2 <- lm(income_midpoint ~ Age + Gender + Race * Education + Hispanic, data = d)
summary(model_c2)
```

```
##
## Call:
## lm(formula = income_midpoint ~ Age + Gender + Race * Education +
##     Hispanic, data = d)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -173848  -43530   -8113   43569  212222
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    15211.93    1476.09  10.306  < 2e-16 ***
## Age                             1240.41      12.48  99.396  < 2e-16 ***
## Genderfemale                  -19009.44     207.28 -91.707  < 2e-16 ***
## RaceBlack                     -17646.00    3556.04  -4.962 6.97e-07 ***
## RaceAsian                      -6382.52    5614.81  -1.137 0.255652
## Raceother                      -7674.62    4168.79  -1.841 0.065627 .
## Educationhigh school           15609.45    1428.49  10.927  < 2e-16 ***
## Educationsome college          29909.14    1398.77  21.382  < 2e-16 ***
## Educationassoc deg             33765.63    1419.23  23.791  < 2e-16 ***
## Educationcollege grad          67843.10    1386.03  48.948  < 2e-16 ***
## Educationadv degree            88416.61    1388.15  63.694  < 2e-16 ***
## HispanicHispanic               -9776.01     358.32 -27.283  < 2e-16 ***
## RaceBlack:Educationhigh school -6343.19    3765.61  -1.685 0.092085 .
## RaceAsian:Educationhigh school -1264.26    6093.28  -0.207 0.835632
## Raceother:Educationhigh school  -323.68    4410.86  -0.073 0.941502
## RaceBlack:Educationsome college -5227.24    3653.06  -1.431 0.152454
## RaceAsian:Educationsome college  7542.49    5819.19   1.296 0.194928
## Raceother:Educationsome college  -857.95    4278.89  -0.201 0.841083
## RaceBlack:Educationassoc deg    -7356.33    3743.22  -1.965 0.049387 *
## RaceAsian:Educationassoc deg     4442.47    5942.53   0.748 0.454719
## Raceother:Educationassoc deg    -1650.66    4384.09  -0.377 0.706536
## RaceBlack:Educationcollege grad -10939.68    3639.86  -3.006 0.002651 **
## RaceAsian:Educationcollege grad  14683.74    5658.16   2.595 0.009456 **
## Raceother:Educationcollege grad  -1910.75    4250.53  -0.450 0.653048
## RaceBlack:Educationadv degree    -6258.66    3631.88  -1.723 0.084843 .
## RaceAsian:Educationadv degree    20879.36    5649.63   3.696 0.000219 ***
## Raceother:Educationadv degree     -675.72    4269.69  -0.158 0.874252
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 58130 on 324904 degrees of freedom
##   (61238 observations deleted due to missingness)
## Multiple R-squared:  0.2272, Adjusted R-squared:  0.2271
## F-statistic:  3674 on 26 and 324904 DF,  p-value: < 2.2e-16
```